note{Both authors contributed equally to this research.}
trovato@corporation.com
1234-5678-9012
G.K.M. Tobin

webmaster@marysville-ohio.com

Institute for Clarity in Documentation
Dublin
Ohio
USA

{Trovato et al.}

Do Not Use This Code Generate the Correct Terms for Your Paper
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1 Introduction

ACM's consolidated article template, introduced in 2017, provides a
consistent LaTeX\ style for use across ACM publications, and incorporates accessibility and metadata-extraction functionality necessary for future Digital Library endeavors. Numerous ACM and
SIG-specific LaTeX\ templates have been examined, and their unique
features incorporated into this single new template.

If you are new to publishing with ACM, this document is a valuable
guide to the process of preparing your work for publication. If you
have published with ACM before, this document provides insight and
instruction into more recent changes to the article template.

The "`acmart`" document class can be used to prepare articles
for any ACM publication — conference or journal, and for any stage
of publication, from review to final "camera-ready" copy, to the
author's own version, with { very} few changes to the source.

2 Template Overview

As noted in the introduction, the "`acmart`" document class can be used to prepare many different kinds of documentation — a double-anonymous initial submission of a full-length technical paper, a
two-page SIGGRAPH Emerging Technologies abstract, a "camera-ready"
journal article, a SIGCHI Extended Abstract, and more — all by selecting the appropriate { template style} and {
template parameters}.

This document will explain the major features of the document class. For further information, the { LaTeX\ User's Guide} is available from
https://www.acm.org/publications/proceedings-template.

2.1 Template Styles

The primary parameter given to the "`acmart`" document class is the { template style} which corresponds to the kind of publication or SIG publishing the work. This parameter is enclosed in square brackets and is a part of the {`documentclass`} command:

```
\documentclass[STYLE]{acmart}
```

Journals use one of three template styles. All but three ACM journals
use the {`acmsmall`} template style:

- {`acmsmall`}: The default journal template style.
- {`acmlarge`}: Used by JOCCH and TAP.
- {`acmtog`}: Used by TOG.

The majority of conference proceedings documentation will use the {`acmconf`} template style.

- {`sigconf`}: The default proceedings template style.
- {`sigchi`}: Used for SIGCHI conference articles.
- {`sigplan`}: Used for SIGPLAN conference articles.

2.2 Template Parameters

In addition to specifying the { template style} to be used in formatting your work, there are a number of { template parameters}
which modify some part of the applied template style. A complete list
of these parameters can be found in the { LaTeX\ User's Guide.}

Frequently-used parameters, or combinations of parameters, include:

- {`anonymous,review`}: Suitable for a "double-anonymous" conference submission. Anonymizes the work and includes line numbers. Use with the `\string\acmSubmissionID` command to print the
submission's unique ID on each page of the work.
- {`authorversion`}: Produces a version of the work suitable for posting by the author.

- {screen}: Produces colored hyperlinks.

This document uses the following string as the first command in the source file:

```
\documentclass[acmtog]{acmart}
```

3 Modifications

Modifying the template — including but not limited to: adjusting margins, typeface sizes, line spacing, paragraph and list definitions, and the use of the `\vspace` command to manually adjust the vertical spacing between elements of your work — is not allowed.

{ Your document will be returned to you for revision if modifications are discovered.}

4 Typefaces

The "acmart" document class requires the use of the "Libertine" typeface family. Your TeX\ installation should include this set of packages. Please do not substitute other typefaces. The "lmodern" and "ltimes" packages should not be used, as they will override the built-in typeface families.

5 Title Information

The title of your work should use capital letters appropriately - https://capitalizemytitle.com/ has useful rules for capitalization. Use the {title} command to define the title of your work. If your work has a subtitle, define it with the {subtitle} command. Do not insert line breaks in your title.

If your title is lengthy, you must define a short version to be used in the page headers, to prevent overlapping text. The title command has a "short title" parameter:

```
\title[short title]{full title}
```

6 Authors and Affiliations

Each author must be defined separately for accurate metadata identification. As an exception, multiple authors may share one affiliation. Authors' names should not be abbreviated; use full first names wherever possible. Include authors' e-mail addresses whenever possible.

Grouping authors' names or e-mail addresses, or providing an "e-mail alias," as shown below, is not acceptable:

```
\author{Brooke Aster, David Mehldau}
\email{dave,judy,steve@university.edu}
\email{firstname.lastname@phillips.org}
```

The authornote and authornotemark commands allow a note to apply to multiple authors — for example, if the first two authors of an article contributed equally to the work.

If your author list is lengthy, you must define a shortened version of the list of authors to be used in the page headers, to prevent overlapping text. The following command should be placed just after the last `\author{}` definition:

```
\renewcommand{\shortauthors}{McCartney, et al.}
```

Omitting this command will force the use of a concatenated list of all of the authors' names, which may result in overlapping text in the page headers.

The article template's documentation, available at https://www.acm.org/publications/proceedings-template, has a complete explanation of these commands and tips for their effective use.

Note that authors' addresses are mandatory for journal articles.

7 Rights Information

Authors of any work published by ACM will need to complete a rights form. Depending on the kind of work, and the rights management choice made by the author, this may be copyright transfer, permission, license, or an OA (open access) agreement.

Regardless of the rights management choice, the author will receive a copy of the completed rights form once it has been submitted. This form contains LaTeX\ commands that must be copied into the source document. When the document source is compiled, these commands and their parameters add formatted text to several areas of the final document:

- the "ACM Reference Format" text on the first page.
- the "rights management" text on the first page.
- the conference information in the page header(s).

Rights information is unique to the work; if you are preparing several works for an event, make sure to use the correct set of commands with each of the works.

{\char'134}	100	Author
{\char'134}	400	For wider tables

searc

The ACM Computing Classification System — https://www.acm.org/publications/class-2012 — is a set of classifiers and concepts that describe the computing discipline. Authors can select entries from this classification system, via https://dl.acm.org/ccs/ccs.cfm, and generate the commands to be included in the LaTeX\ source.

User-defined keywords are a comma-separated list of words and phrases of the authors' choosing, providing a more flexible way of describing the research being presented.

CCS concepts and user-defined keywords are required for for all articles over two pages in length, and are optional for one- and two-page articles (or abstracts).

9 Sectioning Commands

Your work should use standard LaTeX\ sectioning commands: `\section`, `\subsection`, `\subsubsection`, `\paragraph`, and `\subparagraph`. The sectioning levels up to `\subsubsection` should be numbered; do not remove the numbering from the commands.

Simulating a sectioning command by setting the first word or words of a paragraph in boldface or italicized text is { not allowed.}

Below are examples of sectioning commands.

9.1 Subsection

This is a subsection.

9.1.1 Subsubsection

This is a subsubsection.

Paragraph

This is a paragraph.

Subparagraph

This is a subparagraph.

10 Tables

The "`acmart`" document class includes the "`booktabs`" package — https://ctan.org/pkg/booktabs — for preparing high-quality tables.

Table captions are placed { above} the table.

Because tables cannot be split across pages, the best placement for them is typically the top of the page nearest their initial cite. To ensure this proper "floating" placement of tables, use the environment **table** to enclose the table's contents and the table caption. The contents of the table itself must go in the **tabular** environment, to be aligned properly in rows and columns, with the desired horizontal and vertical rules. Again, detailed instructions on **tabular** material are found in the *LaTeX\ User's Guide*.

Immediately following this sentence is the point at which Table ?? is included in the input file; compare the placement of the table here with the table in the printed output of this document.

Table: Frequency of Special Characters

To set a wider table, which takes up the whole width of the page's live area, use the environment **table*** to enclose the table's contents and the table caption. As with a single-column table, this wide table will "float" to a location deemed more desirable. Immediately following this sentence is the point at which Table ?? is included in the input file; again, it is instructive to compare the placement of the table here with the table in the printed output of this document.

Table: Some Typical Commands

Always use midrule to separate table header rows from data rows, and use it only for this purpose. This enables assistive technologies to recognise table headers and support their users in navigating tables more easily.

11 Math Equations

You may want to display math equations in three distinct styles: inline, numbered or non-numbered display. Each of the three are discussed in the next sections.

11.1 Inline (In-text) Equations

A formula that appears in the running text is called an inline or in-text formula. It is produced by the **math** environment, which can be invoked with the usual `{\char'134}begin\,\ldots{\char'134}end` construction or with the short form `\$\,\ldots\$`. You can use any of the symbols and structures, from α to ω, available in LaTeX [1]; this section will simply show a few examples of in-text equations in context. Notice how this equation:

$$\lim_{n\rightarrow \infty}x=0$$

, set here in in-line math style, looks slightly different when set in display style. (See next section).

11.2 Display Equations

A numbered display equation—one set off by vertical space from the text and centered horizontally—is produced by the **equation** environment. An unnumbered display equation is produced by the **displaymath** environment.

Again, in either environment, you can use any of the symbols and structures available in LaTeX\@; this section will just give a couple of examples of display equations in context. First, consider the equation, shown as an inline equation above:

$$\lim_{n\rightarrow \infty}x=0$$

Notice how it is formatted somewhat differently in the **displaymath** environment. Now, we'll enter an unnumbered equation:

$$\sum_{i=0}^{\infty} x + 1$$

and follow it with another numbered equation:

$$\sum_{i=0}^{\infty}x_i=\int_{0}^{\pi+2} f$$

just to demonstrate LaTeX's able handling of numbering.

12 Figures

The "`figure`" environment should be used for figures. One or more images can be placed within a figure. If your figure contains third-party material, you must clearly identify it as such, as shown in the example below.

Figure 1. 1907 Franklin Model D roadster. Photograph by Harris & Ewing, Inc. [Public domain], via Wikimedia Commons. (https://goo.gl/VLCRBB).

Your figures should contain a caption which describes the figure to the reader.

Figure captions are placed { below} the figure.

Every figure should also have a figure description unless it is purely decorative. These descriptions convey what's in the image to someone who cannot see it. They are also used by search engine crawlers for indexing images, and when images cannot be loaded.

A figure description must be unformatted plain text less than 2000 characters long (including spaces). { Figure descriptions should not repeat the figure caption – their purpose is to capture important information that is not already provided in the caption or the main text of the paper.} For figures that convey important and complex new information, a short text description may not be adequate. More complex alternative descriptions can be placed in an appendix and referenced in a short figure description. For example, provide a data table capturing the information in a bar chart, or a structured list representing a graph. For additional information regarding how best to write figure descriptions and why doing this is so important, please see https://www.acm.org/publications/taps/describing-figures/.

12.1 The "Teaser Figure"

A "teaser figure" is an image, or set of images in one figure, that are placed after all author and affiliation information, and before the body of the article, spanning the page. If you wish to have such a
figure in your article, place the command immediately before the `\maketitle` command:

```
\begin{teaserfigure}
  \includegraphics[width=\textwidth]{sampleteaser}
  \caption{figure caption}
  \Description{figure description}
\end{teaserfigure}
```

13 Citations and Bibliographies

The use of {
BibTeX}\ for the preparation and formatting of one's references is strongly recommended. Authors' names should be complete
— use full first names ("Donald E. Knuth") not initials
("D. E. Knuth") — and the salient identifying features of a reference should be included: title, year, volume, number, pages, article DOI, etc.

The bibliography is included in your source document with these two
commands, placed just before the \verb|